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Ramy Gharbi

FinTech & AI Engineer

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Education

Software Engineering Degree, Finance Specialization : Private Higher School of Engineering and Technology 2022 – 2026
Mathematics and Physics Pre engineering Studies : Université Centrale 2021 – 2022

Experience

Freelance Software Engineer **September 2026**
Independent Contractor, Remote
- Built and shipped a luxury e-commerce platform (Next.js 15, TypeScript, Sanity CMS) for a boutique client — server-rendered catalogue with a custom facet-matching & quiz-based product-matching engine, a Tailwind design-token theme, and Framer Motion micro-interactions.

Machine Learning Intern **March – September 2026**
Coral-io Software, Tunisia *MARTA — Final Year Project (PFE)*
- Designed a three-layer algorithmic trading system (Gaussian HMM regime encoder, 5-agent LangGraph deliberation layer, PPO execution agent with CVaR-normalized reward) validated on a 5-asset crypto panel; built a rigorous evaluation framework (walk-forward splits, block-bootstrap CIs, ablation study, 90-debate A/B harness), delivered and root-caused a methodologically sound negative result, then shipped the fix and a live paper-trading dashboard (FastAPI/Next.js, Alpaca, Telegram alerts).

Freelance Software Engineer **November 2025 – April 2026**
Independent Contractor, Bizerte, Tunisia
- Built and deployed a Dockerized full-stack rental platform (FastAPI, Next.js 14, PostgreSQL 16; 22+ tables, 80+ REST endpoints) with JWT auth, Redis token blacklisting, RBAC, and server-side PDF generation; implemented fleet workflows (reservation→rental lifecycle, maintenance alerts, damage reporting) and deployed via docker-compose to DigitalOcean — sold to the client.

Machine Learning Intern **July – August 2025**
Coral-io Software, Tunisia
- Built an end-to-end face clustering pipeline (YOLOv11n-face + FaceNet) and deployed a low-latency FastAPI inference API (ONNX Runtime, Docker) with scalable embedding storage/retrieval (MongoDB + S3-compatible storage).

Machine Learning Intern **July – August 2024**
Société Tunisienne de Banque, Tunisia
- Built an end-to-end stock forecasting pipeline on 13 years of STB data (2010–2022) — feature engineering (RSI, MACD, Bollinger Bands) and model training across ARIMA, LSTM, and ensemble ML models — validated with a leakage-safe walk-forward backtesting framework.

Project

Algorithmic Trading Simulator **December 2025**
ESPRIT *Tunis Stock Exchange*
- Developed backend services for a simulated trading platform (market/limit/stop orders) with portfolio tracking, performance metrics (returns, drawdown, Sharpe ratio), and a modular event-driven backtesting engine using WebSockets for real-time price updates.

Monte Carlo Simulation Lab **November 2025**
Academic Project *GitHub*
- Built a Monte Carlo simulation platform (Python/Flask) for derivatives pricing and risk analysis — stochastic path generation with variance reduction, Black-Scholes/Longstaff-Schwartz option pricing, and risk metrics (VaR, CVaR, Greeks) exposed via a REST API with interactive visualization.

Skills

- **Languages**: Python, Java, SQL, TypeScript, C/C++, R
- **ML/AI Frameworks**: PyTorch, TensorFlow, Scikit-Learn, Hugging Face, LangGraph, Stable-Baselines3, MLflow, FastAPI
- **ML Specialties**: Computer Vision, NLP/LLMs, Reinforcement Learning, Multi-Agent / Agentic AI, Forecasting, Graph Neural Networks
- **FinTech**: Algorithmic Trading & Backtesting, Risk Metrics (VaR, CVaR), Regime Detection (HMM), Volatility Modeling (GARCH), Time Series Analysis, Live Execution (Alpaca)
- **Cloud & Data**: AWS (S3, Lambda, SageMaker), Docker, PostgreSQL, MongoDB, Redis
- **Full-Stack & DevOps**: Next.js, React, Node.js, Git, CI/CD, Agile

Languages

- **Arabic**: Native
- **English**: B2
- **French**: B2

Achievements

- **1st Place**, Code it up
- **1st Place**, Code2Cure
- **4th Place**, AI 4 the blue horizon